

Stay or Switch? Strategic Memory Portability in Competing AI Assistants

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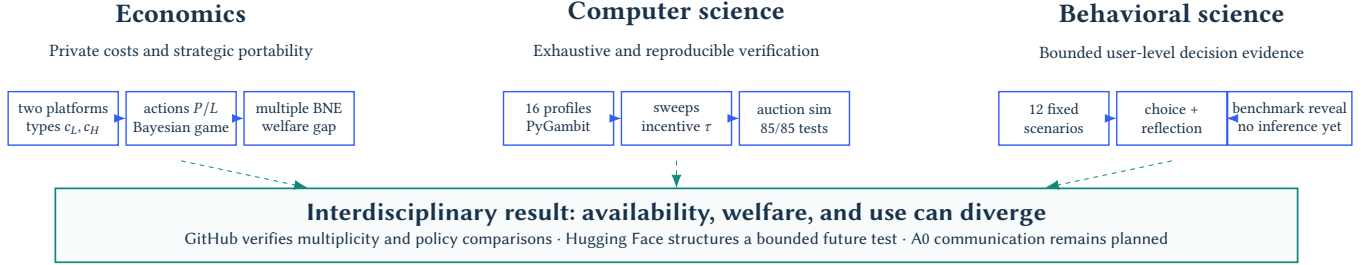


Figure 1: Evidence chain for strategic memory portability. The platform game, computation, welfare/incentive comparisons, and allocation analogy are completed at snapshot f9523b0. The public Stay/Switch artifact structures a separate user test; behavioral inference and the A0 poster remain planned.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** FP10. **Symposium:** Pending course assignment.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Three connected research questions and verified gap

AI-assistant memory can create switching costs and lock-in [1]; portability can reduce that friction, but effects depend on the market [4, 8]. Prior work models strategic compatibility in complementary products [3]. This project instead asks when two platforms with private implementation costs supply memory portability, whether strategic adoption matches a stated collective objective, and whether users would actually switch.

Q1—Economics: Which type-contingent portability choices are Bayesian equilibria, and how do welfare and a verified-portability incentive compare? **Q2—Computation:** Do exhaustive checks, sweeps, and an allocation analogy reproduce the thresholds? **Q3—Behavior:** When a benchmark favors switching, does persistence reflect status quo or omitted trust, privacy, and personalization? The shared question is when portability becomes both supplied and useful. Figure 1 separates completed platform evidence from the planned user test.

2 Answer to Q1: incentives and strategic model

Nature independently draws each platform’s private cost $c_i \in \{c_L, c_H\}$ with $\Pr(c_i = c_L) = p$; platforms simultaneously choose Portable (P) or Locked (L). With rival portability probability q ,

$$EU(P \mid c_i) = 3q - 1 - c_i, \quad EU(L \mid c_i) = q,$$

so P is a weak best response iff $2q - 1 \geq c_i$ [2]. At $c_L = .2, c_H = 1.2, p = .7$, type-level checks yield exactly two pure BNE, (LL, LL) and (PL, PL) , plus a symmetric mixed equilibrium: high types lock and low types choose P with probability $6/7$. Multiplicity is not resolved.

Game theory gives the PL threshold $p \geq (1 + c_L)/2 = .6$. *Social choice* adds the cardinal objective $W = u_A + u_B + U_{user}$: at mobility value $m = 1$, $W_{LL} = 0$, $W_{PL} = 2.17$, and the non-equilibrium comparator $W_{PP} = 4$. *Mechanism design* adds benefit τ , changing the threshold to $p \geq (1 + c_L - \tau)/2$. At $p = .4, \tau = .4$ sustains PL , but LL remains a weak BNE through $\tau = 1.2$: the intervention expands equilibria without selecting one.

3 Answer to Q2: computation, auction comparison, and artifacts

Python enumerates all $4 \times 4 = 16$ pure profiles and checks every type against deviations; an independent PyGambit Harsanyi game returns the same two pure BNE. Integer-built grids sweep $p = .01, \dots, .99, c_L = 0, .05, \dots, 1.15$, and $\tau = 0, .05, \dots, 2.5$. Saved CSV/JSON includes rejected profiles; 85/85 tests pass.

The allocation analogy assigns one primary-assistant slot between two iid $U[0, 1]$ platform values. Portability lowers the assumed reserve from $r = .5$ to $r = .2$. Second price uses truthful bids [9]; first price uses $b(v; r) = (v^2 + r^2)/(2v)$ for $v \geq r$. With seed 20603 and 100,000 common draws, allocation rises from .75193 (analytical .75) to .96004 (.96). Locked/portable unconditional payments are .41788/.36320 (first price) and .41818/.36414 (second price); conditional efficiency is one. The user is not auctioned, reserves are normalized assumptions, and no optimal-auction claim is made [5].

GitHub: snapshot f9523b0. **Notebook:** Phase 3 Colab.

Hugging Face: public Static Space. **A0 poster:** Planned; no result claimed.

4 Answer to Q3: behavior and interdisciplinary integration

The public Static Space crosses hurdles $r = .20, .35$, or $.50$ with benefits $g = .15, .30, .45$, or $.60$. A user decides before seeing the rule, rates switching difficulty, personalization loss, privacy, and trust, then sees the simplified benchmark “Switch iff $g > r$ ” and decides again. Six scenarios predict Stay and six Switch. Status-quo bias evidence motivates the hypothesis of persistence [6], but a Stay choice can instead reflect preferences omitted from $g - r$. The Space has no backend, tracking, persistent store, or cross-browser aggregation; peer summaries use prior plays in the current page session only. It is publicly deployed and technically verified, but no participant-level conclusion is claimed.

A future structured classroom test would compare initial/final decisions and ratings across the fixed scenarios. Persistence concentrated when $g - r > 0$, conditional on measured concerns, would qualify the user-mobility term; sensitivity to disclosure would motivate interface revision. Benchmark alignment would retain the simple rule only as a descriptive classroom benchmark. Either result would leave platform BNE as a separate unit of analysis.

5 Advanced development, real-world impact, and roadmap

Harsanyi’s incomplete-information foundation and the Vickrey–Myerson auction literature discipline the economic claims; reproducible enumeration makes every profile inspectable. The distinctive contribution is an auditable chain from privately costly platform portability, through welfare and incentives, to user mobility—while keeping firm equilibria, simulated allocations, and prospective behavioral evidence separate.

Consider a student or writer deciding whether to move years of AI-assistant context. Users may gain choice; incumbent and entrant platforms face implementation, competition, and security incentives; regulators and institutions need interoperable, privacy-preserving standards. Academia requires identification and replicable evidence, industry requires reliability and security tests, and public institutions require access, accountability, and correction. The next discriminating test is whether portable-memory scenarios increase actual switching relative to no-portability scenarios without raising privacy concern; the present artifact does not answer it.

Table 1: From foundations to an interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	Incomplete-information games; switching costs; auction design.
PS2	Two pure BNE, welfare/incentive comparisons, seeded allocation simulation, public decision artifact.
Symposium	Planned A0 claim and genuine peer challenges; none recorded yet.
Next study	Consent-based scenario comparison with privacy and access safeguards.
2056	Validated interoperable-memory institutions with accountable correction.

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, the aspiration is a tested theory and computational design for interoperable AI memory that aligns platform incentives with user agency. Progress would require multi-market validation, secure standards, mechanism trials, and behavioral evidence; neither awards nor impact are predicted by PS2.

Open Science Statement

Snapshot f9523b0 contains source, two executed notebooks, pinned dependencies, deterministic outputs, tests, and evidence boundaries. From `ps2_memory_portability/`, run `conda run -n cs206-ps2 python -m unittest discover -s tests -v`; the recorded result is 85/85. The Static Space is public; it stores no shared participant dataset. The repository has no new blanket license grant. The poster remains planned.

Statement of Contribution to the UN’s SDGs

No demonstrated SDG impact is claimed. A future secure portability standard could be evaluated against SDG target 9.1’s inclusive-infrastructure objective [7] using successful export/import, accessibility, privacy incidents, and switching completion as indicators. Possible harms include data leakage and exclusion when transferred context is incomplete; audits and user-controlled correction are required.

Submission milestones

Review-ready: September 27, 2026, 11:00 P.M. Beijing time. Symposium and peer evaluation: September 28, 2:45–5:15 P.M., IB 2050. Final revision: September 30, 11:00 P.M. Beijing time. No symposium or peer-review outcome is reported in this review-ready version.

Author Notes

Acknowledgements

Course instructor: Professor Luyao Zhang. The author acknowledges the official COMSCI/ECON 206 PS2 Overleaf Student Template, the open-source Python and LaTeX tools identified in Appendix C, and permitted assistance from OpenAI Codex as disclosed in Appendix A.1. No PS2 peer, discussant, symposium, community-partner, or classroom-auction contribution had been received by this review-ready version; none is invented or attributed. The author remains responsible for the proposal and remaining errors.

Team and individual contributions

Joint contribution statement (solo team; 124 words). This solo project links a static Bayesian game of privately costly memory portability to three inspectable extensions: a cardinal social-welfare comparator, a verified-portability incentive, and a reserve-based platform-allocation analogy. Its contribution is the separation of three questions often conflated in portability debates: whether firms strategically supply portability, whether modeled outcomes rank differently under a stated collective objective, and whether users switch when technical friction falls. Reproducible enumeration, PyGambit agreement, deterministic sweeps, seeded auction simulation, and 85 automated tests verify the formal claims. A public Static Space implements twelve hypothetical Stay/Switch scenarios while explicitly providing no persistent participant dataset. The paper therefore offers an integrated, falsifiable research design and public artifacts, not an empirical claim about firms, users, welfare, or status-quo bias.

Yiqiao Liu (Mickey), sole and first author. Liu selected the memory-portability question; approved the private-cost Bayesian structure, normalized payoffs, welfare objective, verified-portability intervention, auction mapping, and behavioral evidence boundary; and is accountable for interpreting every result. Inspectable outputs are in `ps2_memory_portability/`, especially the two notebooks, `outputs/`, the public Space, and this paper source. Verification includes direct/PyGambit agreement, analytical thresholds, scenario parity, public deployment checks, 85/85 tests, and LaTeX/PDF audits. The official team code is FP10. The next responsibility is to confirm the symposium session, create the A0 poster from its official template, obtain genuine September 28 reviews, and record Keep/Revise/Investigate decisions without delegating personal reflection to AI.

References

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A Technical details and reproducibility

A.1 Model, timing, and type-level verification

Platforms A, B independently draw $c_i \in \{.2, 1.2\}$ with $\Pr(c_i = .2) = .7$, observe only their own type, and simultaneously choose P or L . Normalized own payoff is $2 - c_i$ under (P, P) , $-1 - c_i$ under (P, L) , 1 under (L, P) , and 0 under (L, L) . Thus

$$EU(P | c) = 3q - 1 - c, \quad EU(L | c) = q,$$

$$P \in BR(c) \iff 2q - 1 \geq c.$$

A pure strategy is a complete type map in $\{LL, PL, LP, PP\}$. The checker evaluates every type of both players in all 16 profiles; zero-probability endpoints are excluded from the full-support sweep.

Table 2: Predetermined parameters and evidence status.

Component	Value and status
Platform benchmark	$c_L = .2, c_H = 1.2, p = .7$; normalized, not estimated.
Pure BNE	$(LL, LL), (PL, PL)$; no asymmetric pure BNE at benchmark.
Mixed candidate	High types lock; low types use P with $x^* = 6/7$.
Social choice	$W = u_A + u_B + U_{user}$; cardinal comparability assumed.
Mechanism	$EU_\tau(P c) = 3q - 1 - c + \tau$; mechanism cost unmodeled.
Auction	$N = 100,000$, seed 20603, iid $U[0, 1]$, reserves $.5/.2$.
Behavior	12 fixed scenarios; zero participant-level results.

A.2 Language-independent verification procedure

- (1) Load the predetermined cost, prior, mobility, incentive, reserve, and seed values.
- (2) For each ordered pair of type-contingent strategies, compute the rival-induced q ; for each positive-probability type, compare the prescribed action with its deviation. Save every accepted and rejected profile.
- (3) Cross-check benchmark pure BNE with a separately constructed PyGambit Harsanyi game; fail if sets differ.
- (4) Sweep integer-built p, c_L, τ, m grids; compare classifications with analytical thresholds before plotting.

- (5) Draw one seeded valuation sample; reuse it for first-/second-price and locked/portable conditions; compare simulated $1 - r^2$, payment, utility, and efficiency.
- (6) Run all tests, execute notebooks from clean kernels, validate the behavioral catalog, and compare paper claims with saved JSON/CSV fields.

A.3 Verified outputs and limitations

At $m = 1$, $W_{LL} = 0$, $W_{PL} = 2.17$, and $W_{PP} = 4$; PP is a comparator, not a Phase 2 BNE. For symmetric PL , $p \geq (1 + c_L - \tau)/2$, while the high type locks only when $\tau \leq 1 + c_H - 2p$. At $p = .4$, the minimum τ for weak PL is $.4$. At $p = .7$, LL remains weak through $\tau = 1.2$, and PP first appears weakly at $\tau = .2$. No welfare-optimal τ is claimed.

The auction’s analytical allocation probability is $1 - r^2$. The saved simulation gives .75193 at $r = .5$ and .96004 at $r = .2$, with absolute errors .00193 and .00004. The same valuation draws are used throughout; 400,000 raw in-memory rows are not committed, and a deterministic 400-row sample is retained. The application does not auction a person, calibrate AI markets, or implement a Myerson-optimal mechanism.

The recorded environment is Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3, matplotlib 3.10.8, PyGambit 16.7.0, nbformat 5.11.1, nbclient 0.11.0, and ipykernel 7.3.0. Reproduce from repository snapshot f9523b0:

```
cd ps2_memory_portability
conda run -n cs206-ps2 python scripts/run_phase2.py
conda run -n cs206-ps2 python scripts/run_phase3.py
conda run -n cs206-ps2 python -m unittest \
discover -s tests -v
```

Expected test result: 85 passed, 0 skipped, 0 failures, 0 errors. The two tracked notebooks are executed artifacts, not substitutes for the source modules.

The public Static Space revision is 9ffb3ca (full revision recorded in the audit). It has no backend, analytics, cookies, or persistent browser storage; refreshing or closing clears its page-session aggregate. The review-ready A0 poster does not yet exist and is not misreported as an output.

A.4 AI-use disclosure

OpenAI Codex (GPT-5 family, September 23–26, 2026) assisted with repository inspection, Python implementation and debugging, test design and execution, notebook and visualization construction, literature-source checking, the Gradio and Static Space artifacts, LaTeX drafting, compilation, and evidence/parity audits. Significant instructions required preserving the research question and official template, separating actual from planned evidence, and prohibiting fabricated author, peer, auction, symposium, and participant records. Suggestions that would overstate empirical evidence, infer missing metadata, or treat the user as literally auctioned were rejected. The author supplied or approved the research direction and remains responsible for every modeling choice, citation, interpretation, and final submission. Automated tests and artifact comparisons are recorded; personal reflection, peer review, and final human approval must be completed by the author.

B Cumulative Intellectual Development

PS1 studied costly information acquisition before a collective AI decision: two agents chose Research or Skip, producing a free-riding benchmark and a reproducible NashPy check. PS2 does not claim topic continuity. It retains the methodological lesson that strategic interdependence, solution concept, computation, and behavioral evidence must be separated, then applies it to a new shared question: privately costly portability between competing AI assistants.

Table 3: Dated development record; no peer-review cause is inferred.

Date / evidence		Revision	Intellectual consequence
PS1	final, Sept. 2026	Costly Re-search/Skip benchmark with future behavior test	Established model-first and actual/planned evidence discipline.
Sept. 24, 796bee4	commit	Private-cost P/L Bayesian game	Made implementation cost private and BNE the correct solution concept.
Sept. 24, 181c840		Added welfare, τ , and auction comparison	Exposed equilibrium–welfare tension and a bounded policy lever.
Sept. 25, 7a2e02e		Verified literature claims	Narrowed compatibility, portability, and status-quo claims to supported analogies.
Sept. 26, f9523b0		Public Static Space verified	Separated technical deployment from participant evidence.

The author initially focused on whether portability is supplied. The welfare comparison showed that PP can rank highest without being a benchmark equilibrium; the mechanism sweep showed that an incentive can add portability equilibria while lock-in persists; and the behavioral design showed that technical portability does not logically imply user switching. The current integrated claim is therefore conditional: platform adoption, collective desirability, feasible allocation, and user response are distinct propositions requiring distinct evidence.

C Open-source and three-lens inspiration

Table 4: Software/source record and design consequence.

Source / tool	Version / license	Capability, limit, consequence
Python ecosystem	Versions in App. A; upstream licenses apply	Deterministic analysis; package behavior is environment-dependent.
PyGambit	16.7.0; GPL-2.0-or-later	Independent pure-BNE cross-check; direct checker remains authoritative.
Gradio	6.28.0; upstream license	Local prototype; free hosted compute was unavailable, prompting static conversion.
HTML/CSS/JS	Browser-native	Public zero-build artifact; no shared response store.
ACM acmart	Official course-bundled class	Required paper layout; content does not imply ACM publication.
Course repository	Public; no new blanket license grant	Inspectable evidence; public availability is not additional reuse permission.

The three lenses produce different decisions. Game theory asks which type-contingent strategies survive unilateral deviation. Social choice adds users and a declared collective objective, revealing a contested cardinal-comparability assumption. Mechanism design changes the action payoff through τ , then tests adoption and equilibrium-selection limits. None of these establishes observed firm conduct. The auction comparison tests a separate allocation analogy; the behavioral Space tests a separate user-level decision rule.

Literature is used only for the general mechanisms it supports: switching costs and lock-in [1], strategic compatibility [3], setting-specific portability competition [8], security trade-offs [4], status-quo motivation [6], and auction foundations [5, 9]. Their parameters and outcomes are not transferred to AI assistants.

D Review response and revision record

This document is the review-ready v1 due September 27, before the September 28 symposium. No PS2 peer review, discussant comment, instructor comment, classroom-auction observation, or symposium question had been received when it was compiled. Therefore no substantive Keep/Revise/Investigate decision is attributed to another person. The prior PS1 record concerns a different topic and is historical development evidence, not direct review of this proposal.

Table 5: Review-ready status; empty evidence is reported, not filled.

Required record	Current status	Next verified action
Review-ready v1	Complete	Retain PDF, ZIP, commit, and audit as baseline.
Two peer reviews	Not yet received	After Sept. 28, record named/dated forms and exact claims.
Discussant / instructor	None received for PS2	Record only comments actually delivered.
Author decision	Pending evidence	Classify each consequential comment Keep, Revise, or Investigate.
Rerun/final v2	Not yet applicable	Rerun affected test and cite changed file/location for Sept. 30.

The pre-review parity audit did produce author-side corrections: it preserved the two pure-BNE wording, labeled *PP* a welfare comparator, separated Static page-session aggregation from participant evidence, and labeled the poster and classroom observations as pending. These are verification edits, not peer feedback.

E Structured abstract and cumulative proposal map

Table 6: Appendix E structured abstract; actual and planned evidence remain separate.

Proposal element	Current statement / change
Background	AI assistants accumulate user-specific memory. Switching costs can create lock-in, while firms may make compatibility choices strategically. Existing work does not supply this project's static private-implementation-cost Bayesian result or AI-user evidence.
Motivation and application	A student or writer may want to move long-lived context between assistants. Users, incumbent and entrant platforms, regulators, and institutions face benefits from choice and risks from privacy, incomplete transfer, security, and unequal access.
Three connected questions	Economics asks when portability is a BNE, how it ranks under a collective objective, and how τ changes incentives. Computation asks whether exhaustive checks and allocation simulations reproduce thresholds. Behavior asks whether users switch when $g > r$. Together: when is portability supplied and practically useful for competitive mobility?
Method and model assumptions	Two privately informed platforms simultaneously choose P/L ; $c_L = .2$, $c_H = 1.2$, $p = .7$. Direct enumeration and PyGambit check BNE; deterministic sweeps test welfare and incentive thresholds; a seeded two-bidder allocation analogy compares first and second price at reserves $.5/.2$; twelve fixed scenarios structure future user evidence.
Results or expected results	Actual: pure BNE (LL, LL) , (PL, PL) ; low-type mix $6/7$; $W_{PP} > W_{PL} > W_{LL}$ on the tested m grid; τ expands portability equilibria without selecting one; allocation $.75193/.96004$; 85/85 tests; public Static Space. Planned: participant, classroom-auction, poster, symposium, and causal evidence.
Intellectual merits	The project shows that strategic availability, modeled collective value, feasible allocation, and user adoption are distinct claims. Combining analytical thresholds, exhaustive computation, and a bounded behavioral design prevents one evidence type from standing in for another.
Practical impacts	Potential value is user mobility and contestability; risks include privacy loss, partial transfer, manipulation, and exclusion. No impact is demonstrated. Future evaluation should measure successful transfer, switching, access, and incidents, with user-controlled correction and independent audit.
Feedback received and revision made	No PS2 peer or symposium feedback had been received in review-ready v1. The author-side parity audit narrowed claims, preserved Gradio failure history, recorded successful Static deployment and team code FP10, and marked the session, poster, classroom play, and future review as pending. The prior PS1 topic is preserved as methodological history, not evidence for PS2.

The prior PS1 structured abstract concerned costly information acquisition before collective decisions. Its retained methodological lesson is explicit separation of formal prediction, computation, and uncollected behavior; its substantive results are not reused as memory-portability evidence.

F PS2 extension record: auction, linked artifacts, poster, and symposium

F.1 Auction laboratory record

The completed evidence is a seeded computational benchmark, not a classroom play. Each treatment uses two iid $U[0, 1]$ private platform values, one primary-assistant slot, one sealed bid per platform, Platform A tie-breaking, a reserve, allocation to the highest eligible bid, mechanism-specific payment, and termination after one allocation decision.

Table 7: Completed simulation treatments; $N = 100,000$, seed 20603.

	r	Alloc.	Pay.	Eff.
First, locked	.50	.75193	.41788	1.00
First, portable	.20	.96004	.36320	1.00
Second, locked	.50	.75193	.41818	1.00
Second, portable	.20	.96004	.36414	1.00

First-price eligible bids follow $b(v; r) = (v^2 + r^2)/(2v)$; second-price bids equal value. “Eff.” is conditional efficiency given allocation. Utilities and conditional payments are preserved in outputs/auction_summary.csv. **Classroom record:** no game URL, assigned value/signal, planned bid, actual bid/outcome, payment, utility, revenue, behavioral observation, or resulting revision was available. Those fields must be completed only after genuine play.

F.2 Linked-artifact parity

Artifact	Verified parity status
GitHub	Snapshot f9523b0; same private-cost game, inputs, formulas, outputs, 85 tests, and limitations.
Hugging Face	Same user-level Stay/Switch question, 12 scenarios, $g > r$ rule, no participant conclusion, page-session privacy boundary.
A0 poster	Not yet created. It must reuse the shared question, actual results, real-world case, limits, instructor, acknowledgements, and artifact links; no parity claim is made yet.

F.3 Symposium and practical-impact record

This review-ready package predates the September 28 symposium. No presentation version, peer-evaluation form, question, or Keep/Revise/Investigate response exists yet. For the concrete migration case, affected parties are users, incumbent and entrant assistants, institutions procuring AI, and regulators. Potential benefit is safer competitive mobility; risks are leakage, degraded context, strategic misrepresentation, and inaccessible transfer. Missing evidence includes real implementation costs, security performance, transfer fidelity, switching choices, and distributional effects. Proposed

correction routes are user-controlled export/import, data minimization, provenance logs, deletion and rollback, accessibility testing, complaint procedures, and independent audit. The September 30 record must identify every actual change and rerun its affected check.